LINUX NOTES

Beginner → DevOps | Remember Boxes + Easy Diagrams

Study these like handwritten class notes. Read the concept, look at the diagram, then practice the command.

> REMEMBER
> Study trick: Don't memorize every command at once. Understand what the command does, then use it in the terminal.

1. What is Linux?

• Linux is an open-source operating system kernel. Ubuntu, Debian, Fedora and RHEL are Linux distributions.

• Linux is heavily used in servers, cloud computing and DevOps because it is stable, scriptable and command-line friendly.

User → Shell → Kernel → Hardware
↓
Applications / Commands

> REMEMBER
> Think: Shell is the translator between you and the operating system.

2. Linux File System

• / is the root directory. Everything starts from it.

• /home stores normal users' personal files. /etc stores configuration. /var commonly stores logs and changing data. /tmp stores temporary files.

/
├── home/ → user files
├── etc/ → configuration
├── var/ → logs / changing data
├── tmp/ → temporary files
└── usr/ → programs / libraries

> REMEMBER
> Remember: / means the top of the Linux file system, not the Windows C: drive.

3. Navigation & Basic Commands

• pwd shows where you are. ls shows what is there. cd changes your location.

• mkdir creates a directory. touch creates an empty file. cp copies. mv moves/renames. rm removes.

pwd → Where am I?
ls → What is here?
cd → Move
mkdir / touch → Create
cp / mv → Copy / Move
rm → Delete

> REMEMBER
> ⚠ Be careful with rm -r. It can delete a directory and its contents.

4. Files, Search & Pipes

• cat displays a file. less reads large files. head/tail show beginning/end. grep searches text. find searches for files.

• A pipe | sends the output of one command into another command.

command 1 → output → | → command 2
ps aux | grep nginx
process list filter nginx

> REMEMBER
> Remember: | means “take this output and give it to the next command.”

5. Linux Permissions

• Permissions are r = read, w = write, x = execute. They apply to user, group, others.

• chmod 755 script.sh gives owner rwx and group/others r-x. chown changes ownership.

USER GROUP OTHERS
rwx r-x r--
r = 4 w = 2 x = 1
755 = 7(rwx) + 5(r-x) + 5(r-x)

> REMEMBER
> Easy memory: r=4, w=2, x=1. Add the numbers for each permission group.

6. Users, Groups & sudo

• whoami shows the current user. id shows IDs and groups. sudo runs an authorized command with elevated privileges.

• useradd creates a user. groupadd creates a group. usermod -aG group user adds a user to a group.

User → belongs to → Group
↓ ↓

permissions shared access
↓
sudo → elevated action
REMEMBER □ Root = superuser. Use sudo carefully.

7. Processes & Services

• A process is a running instance of a program. ps aux lists processes and top shows them live.

• systemd commonly manages services: systemctl status/start/stop/restart.

Program → started → Process
↓
PID number
↓
ps / top / kill
Service → systemctl → start/stop/status
REMEMBER □ Remember: PID = Process ID.

8. Networking

• ip addr shows interfaces/IPs. ping tests reachability. ss -tuln shows listening sockets. curl makes HTTP requests. ssh gives secure remote access.

• Common ports: 22 SSH, 80 HTTP, 443 HTTPS.

Your PC ── network ── Server
| |
ip addr IP address
|
ssh / curl
|
Service
REMEMBER □ Think: IP = address, Port = door, Service = application behind the door.

9. Packages on Ubuntu

• sudo apt update refreshes package information. sudo apt upgrade upgrades installed packages. sudo apt install nginx installs software.

Repository → apt → Package

↓
Installed
↓
Application
REMEMBER □ update gets the latest package information. upgrade installs newer versions.

10. Bash & Shell Scripting

• Bash is a common shell. A script is a file containing commands that can be executed automatically.

• Common ideas: variables, if/else, loops, exit status and command substitution.

script.sh
↓
#!/bin/bash
↓
variables / conditions / loops
↓
commands → result
REMEMBER □ First line often used: #!/bin/bash. Then use chmod +x script.sh and ./script.sh.

11. Logs & Troubleshooting

• Logs help you understand what happened. journalctl reads systemd journal logs. journalctl -u service filters a service.

• Basic troubleshooting: check status → read logs → check config → check ports/network → fix → test again.

Problem
↓
Service status
↓
Logs
↓
Config / Port / Network
↓
Fix → Test

REMEMBER Don't restart blindly. First find the reason for the problem.

12. DevOps Linux Command Map

- Navigation: pwd, ls, cd
- Files: mkdir, touch, cp, mv, rm
- Text: cat, less, head, tail, grep
- Search: find
- Permissions: chmod, chown
- Processes: ps, top, kill
- Services: systemctl
- Network: ip, ping, ss, curl, ssh
- Packages: apt
- Automation: bash

Linux
├─ Files
├─ Permissions
├─ Processes
├─ Services
├─ Network
├─ Packages
└─ Bash

REMEMBER DevOps uses Linux as a foundation. Learn the concepts first, then commands.

13. Practice Tasks

- 1. Create a devops directory and three files.
- 2. Create a Bash script and make it executable.
- 3. Create a user and group, then add the user to the group.
- 4. Install nginx and check its status.
- 5. Use grep and tail -f to inspect a log.
- 6. Use ss -tuln to see listening ports.
- 7. Use ps and kill to practice process management.

Learn → Run → Observe → Make a mistake → Fix → Repeat

REMEMBER Best way to learn Linux: open Ubuntu and type the commands yourself.

□ ONE-PAGE REVISION

COMMAND MEMORY pwd = location | ls = list | cd = change directory | mkdir = create directory | touch = create file
FILE MEMORY cp = copy | mv = move/rename | rm = remove | cat = display | grep = search
PERMISSION MEMORY r=4, w=2, x=1 | 755 = rwx r-x r-x | chmod = permissions | chown = ownership
PROCESS MEMORY ps = processes | top = live processes | kill = terminate | systemctl = services
NETWORK MEMORY ip = network info | ping = reachability | ss = sockets/ports | ssh = remote access | curl = HTTP request
APT MEMORY apt update = refresh package information | apt upgrade = upgrade installed packages | apt install = install software

□ Mini DevOps Flow

Developer → Git → Linux Server → Application
↓
Logs + Monitoring
↓
Troubleshooting
FINAL REMEMBER If you can explain the diagram in your own words and perform the commands in Ubuntu, your Linux foundation is becoming strong.